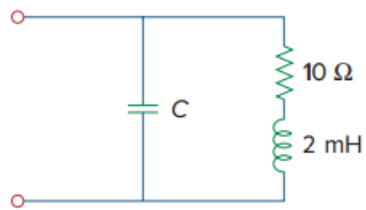


Problem

Figure 9.91 shows a series combination of an inductance and a resistance. If it is desired to connect a capacitor in parallel with the series combination such that the net impedance is resistive at 10 kHz, what is the required value of C ?

Figure 9.91:



Step-by-step solution

Step 1 of 6

Refer to Figure 9.91 in the text book.

The frequency, f is given as 10 kHz.

Calculate the angular frequency, ω .

$$\omega = 2\pi f$$

Substitute 10 kHz for f .

$$\begin{aligned}\omega &= (2\pi)(10 \times 10^3) \\ &= 62831.853 \text{ rad/s}\end{aligned}$$

Step 2 of 6

Assume the value of the capacitor to be C and the angular frequency to be ω rad/s.

Write the expression for the impedance offered by the capacitor, Z_1 .

$$Z_1 = \frac{1}{j\omega C}$$

Step 3 of 6

Calculate the impedance of the series combination of the inductor and the resistor.

$$Z_2 = R + j\omega L$$

Substitute 10Ω for R and 2 mH for L .

$$\begin{aligned}Z_2 &= 10 + j\omega(2 \times 10^{-3}) \\ &= 10 + j0.002\omega\end{aligned}$$

Step 4 of 6

Calculate the equivalent impedance Z_{eq} , as seen from the opened terminals.

$$Z_{eq} = Z_1 \parallel Z_2$$

$$Z_{eq} = \frac{1}{j\omega C} \parallel (10 + j0.002\omega)$$

$$Z_{eq} = \frac{\left(\frac{1}{j\omega C}\right)(10 + j0.002\omega)}{\left(\frac{1}{j\omega C}\right) + (10 + j0.002\omega)}$$

$$Z_{eq} = \frac{10 + j0.002\omega}{j\omega C + \frac{j\omega C(10 + j0.002\omega)}{j\omega C + (10 + j0.002\omega)}}$$

Thus,

$$Z_{eq} = \frac{10 + j0.002\omega}{j\omega 10C - \omega^2 0.002C + 1}$$

Step 5 of 6

Express the equivalent impedance as a sum of real and imaginary parts.

$$\begin{aligned} Z_{eq} &= \frac{10 + j0.002\omega}{(1 - \omega^2 0.002C) + j10\omega C} \times \frac{(1 - \omega^2 0.002C) - j10\omega C}{(1 - \omega^2 0.002C) - j10\omega C} \\ &= \frac{(10 + j0.002\omega)((1 - \omega^2 0.002C) - j10\omega C)}{(1 - \omega^2 0.002C)^2 + (10\omega C)^2} \\ &= \frac{(10 + j0.002\omega)(1 - \omega^2 0.002C) - j100\omega C + 0.02\omega^2 C}{(1 - \omega^2 0.002C)^2 + (10\omega C)^2} \\ &= \frac{10 + j0.002\omega - \omega^2 0.02C - j\omega^3 (4 \times 10^{-6})C - j100\omega C + 0.02\omega^2 C}{(1 - \omega^2 0.002C)^2 + (10\omega C)^2} \end{aligned}$$

Thus,

$$\begin{aligned} Z_{eq} &= \frac{10 + j0.002\omega - \omega^2 0.02C - j\omega^3 (4 \times 10^{-6})C - j100\omega C + 0.02\omega^2 C}{(1 - \omega^2 0.002C)^2 + (10\omega C)^2} \\ &= \frac{10}{(1 - \omega^2 0.002C)^2 + (10\omega C)^2} + j \frac{(0.002\omega - \omega^3 (4 \times 10^{-6})C - 100\omega C)}{(1 - \omega^2 0.002C)^2 + (10\omega C)^2} \end{aligned}$$

Step 6 of 6

For impedance to be pure resistive, the imaginary part should be zero.

$$0.002\omega - \omega^3(4 \times 10^{-6})C - 100\omega C = 0$$

$$C = \frac{0.002\omega}{\omega^3(4 \times 10^{-6}) + 100\omega}$$

$$C = \frac{0.002}{\omega^2(4 \times 10^{-6}) + 100}$$

Substitute 62831.853 rad/s for ω .

$$\begin{aligned} C &= \frac{0.002}{(62831.853)^2(4 \times 10^{-6}) + 100} \\ &= \frac{0.002}{15891.367} \\ &= 1.258 \times 10^{-7} \text{ F} \\ &= 0.126 \mu\text{F} \end{aligned}$$

Note: The answer given in the book do not considers the factor of 2π due to which the answer comes out to be wrong.

Therefore, the correct value of the capacitor required should be $\boxed{0.126 \mu\text{F}}$.